An Empirical Benchmark of QAOA in Cardinality-Constrained Portfolio Optimization

Roan Cedric Campo
De La Salle University
Manila, Philippines
roan_campo@dlsu.edu.ph

Enzo Rafael Chan
De La Salle University
Manila, Philippines
enzo_rafael_chan@dlsu.edu.ph

***Abstract*—Modern Portfolio Theory relies on the Markowitz Mean-Variance model to optimize asset allocation. While classical convex optimization solves the unconstrained model efficiently, introducing strict discrete cardinality limits transforms the problem into an NP-hard challenge. This paper explores the application of the Quantum Approximate Optimization Algorithm (QAOA) to solve the cardinality-constrained Markowitz model using a D-bit integer weighting architecture. By relaxing the discrete constraint into a quadratic penalty term, the objective is translated into a Quadratic Unconstrained Binary Optimization (QUBO) formulation. We conduct an exhaustive parameterized grid search—dynamically tuning risk, reward, and cardinality penalty weights (λ_{card})—to benchmark the resulting portfolio outputs against a rigorous hierarchy of classical exact (MIQP) and heuristic baselines. Our empirical analysis reveals a critical computational trade-off: while heavy λ_{card} penalties successfully force convergence on target cardinality, the exponential coefficient scaling inherent to continuous-to-discrete mappings simultaneously induces severe energy landscape "ruggedness" and exacerbates barren plateau phenomena. Evaluated via high-fidelity classical simulators to isolate algorithmic bottlenecks from physical hardware noise, this study provides a rigorous quantitative analysis of the scaling trends, fidelity ceilings, and hyperparameter sensitivities characteristic of standard penalty-based QAOA implementations in computational finance.**

***Index Terms*—QAOA, Quantum Computational Finance, Portfolio Optimization, Cardinality Constraints, QUBO, Algorithmic Scaling, NISQ, Benchmarking**

I. Introduction

Discrete cardinality-constrained portfolio optimizationpresents a fundamental computational challenge in quantitative finance. The requirement to select exactly K assets from an n-asset universe while simultaneously optimizing a risk-adjusted return objective transforms the otherwise tractable Markowitz Mean-Variance model into an NP-hard combinatorial problem. As institutional portfolios scale to hundreds of assets, the search space grows combinatorially, rendering exact classical enumeration infeasible and motivating the exploration of novel computational paradigms, including quantum optimization.

Modern Portfolio Theory, foundationalized by Harry Markowitz in 1952 [1], revolutionized financial economics by framing asset allocation as a mathematical optimization problem. The core objective of the Markowitz Mean-Variance model is to maximize expected portfolio returns for a given level of market risk, or conversely, to minimize risk for a target expected return. In its purest, unconstrained form, this is a convex quadratic programming problem that classical algorithms can solve efficiently.

However, real-world portfolio management necessitates the introduction of complex, non-convex constraints. Practitioners must account for cardinality (limiting the total number of assets to minimize transaction costs and management overhead) and specific risk-reward mandates. The integration of these discrete constraints transforms the classical Markowitz model from a highly tractable convex optimization task into an NP-hard problem. As the asset universe expands, classical solvers struggle to scale, often relying on heuristic or metaheuristic approximations that cannot guarantee an optimal energetic minimum [2].

Quantum computing presents a novel paradigm for overcoming these computational bottlenecks. By translating the constrained portfolio optimization problem into a Quadratic Unconstrained Binary Optimization (QUBO) formulation, the financial model can be mapped directly to an Ising Hamiltonian. In this framework, the quantum computer evaluates the problem as a complex energy landscape, where the lowest energy state corresponds to the optimal asset allocation.

This paper explores the application of the Quantum Approximate Optimization Algorithm (QAOA) [3], which is a hybrid quantum-classical algorithm specifically designed for Noisy Intermediate-Scale Quantum (NISQ) devices, to solve the highly constrained Markowitz model. We benchmark the QAOA-generated portfolios against a classical baseline, evaluating key performance metrics including the Sharpe ratio, execution time, and algorithmic ruggedness. Furthermore, we analyze the probabilistic nature of the quantum outputs, investigating how inherent hardware noise within the NISQ era influences the validity and probability distributions of the optimal financial state.

II. Related Works

The intersection of quantum computing and financial mathematics has seen rapid development, particularly in translating the Markowitz Mean-Variance model into quantum-native formats. While classical optimization efficiently solves the continuous formulation of the portfolio problem, the introduction of discrete constraints, such as cardinality limits, renders the problem NP-hard, motivating the exploration of quantum algorithms [4].

Recent literature has explored the conversion of constrained portfolio optimization into Quadratic Unconstrained Binary Optimization (QUBO) problems. A comprehensive benchmark by Stopfer et al. compared QAOA and quantum annealing against classical Mixed-Integer Programming (MIP) solvers, highlighting that while classical solvers remain highly efficient for standard discrete formulations, quantum algorithms demonstrate significant scaling potential as asset universes grow [5]. Their work underscores the necessity of optimal penalty parameter (λ) tuning when enforcing hard constraints in a QUBO landscape.

To manage the complexity of specific practitioner constraints, several novel QAOA variations have been proposed, including constraint-preserving formulations utilizing XY-Mixers [6]. However, standard implementations still rely heavily on penalty terms that can degrade quantum circuit performance and create rugged energy landscapes [7]. This paper builds upon this identified gap by performing a rigorous empirical analysis of the standard QUBO penalty method, benchmarking the impact of hyperparameter tuning on algorithmic stability and measurement fidelity in NISQ environments.

III. Methodology

The execution of this study follows a dual-track methodology: establishing a classical upper bound portfolio using standard convex optimization, and subsequently translating the constrained problem into a quantum-native format for execution via the Quantum Approximate Optimization Algorithm (QAOA).

A. Software Pipeline and Classical Baseline

This study relies on two third-party open-source libraries, neither of which was developed by the authors. Full attribution is provided below.

yfinance (R. Aroussi [8]) is a Python interface to the Yahoo Finance API, providing programmatic access to historical adjusted closing prices, dividend records, and corporate action data. In this study, `yfinance` was used to retrieve five years of daily adjusted closing prices for the 16 S&P 500 constituents comprising the asset universe. To preserve temporal integrity, assets with incomplete trading records within the selected window were excluded via `DataFrame.dropna()` rather than imputed through forward-filling, which would introduce look-ahead bias into the covariance estimate.

PyPortfolioOpt (R. Martin [9]) is a portfolio optimization library implementing Modern Portfolio Theory constructs. Its `EfficientFrontier` class constructs the mean-variance efficient frontier by solving a convex quadratic program via the OSQP (Operator Splitting Quadratic Program) solver. Given the sample covariance matrix $\Sigma \in \mathbb{R}^{n\times n}$ and the expected return vector $\boldsymbol{\mu} \in \mathbb{R}^n$ derived from historical price data, `PyPortfolioOpt` computes the maximum Sharpe ratio portfolio subject to long-only bounds ($w_i \geq 0$) and full investment ($\mathbf{1}^\top \mathbf{w} = 1$). This continuous solution serves as the theoretical upper bound—the *Continuous Ceiling*—against which all discrete solver outputs are benchmarked.

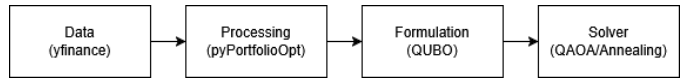

Fig. 1: Data processing and algorithmic execution pipeline. Market data is acquired via `yfinance`, optimized classically via `PyPortfolioOpt`, formulated into a QUBO, and executed across quantum and analog heuristic solvers.

Figure 1 illustrates the end-to-end pipeline, highlighting the transformation from financial data acquisition to QUBO formulation and solver evaluation.

B. Problem Specification and Proper QUBO Formulation

We define the portfolio over an N-asset universe. Rather than restricting allocations to strict binary inclusion (1-bit), our architecture supports integer-weighted allocations using a D-bit depth system. The fractional weight of asset i is represented by a sequence of D binary variables (qubits) denoted as $q_{i,k} \in \{0,1\}$ for $k = 0, \ldots, D-1$.

The total integer allocation units assigned to asset i is evaluated via the binary expansion:

$$y_i = \sum_{k=0}^{D-1} 2^k q_{i,k} \tag{1}$$

The normalized portfolio weight is thus $w_i = \frac{y_i}{R_{\max}}$, where $R_{\max} = 2^D - 1$ is the maximum allocation per asset. The discrete state space expands exponentially; for an N-asset universe at D-bit depth, the system requires $N \times D$ logical qubits.

The core objective is to minimize risk (Σ) and maximize expected returns (μ), subject to a strict cardinality constraint restricting the total number of allocated units across the portfolio to exactly K_{target}. The classical objective is augmented by a squared penalty term to cast the problem as a Quadratic Unconstrained Binary Optimization (QUBO) model:

$$\text{Cost} = \lambda_{\text{risk}} \left(w^T \Sigma w\right) - \lambda_{\text{reward}} \left(\mu^T w\right) + \lambda_{\text{card}} \left(\sum_{i=1}^{N} y_i - K_{\text{target}}\right)^2 \tag{2}$$

To construct the corresponding QUBO Hamiltonian $H(q) = \sum Q_{uu} q_u + \sum_{u<v} Q_{uv} q_u q_v$, where indices u, v flatten the (i,k) mapping, we algebraically expand the integer components. Noting the boolean identity $q_{i,k}^2 = q_{i,k}$, the penalty expansion introduces highly dense quadratic interactions. Specifically, the cardinality penalty term $\lambda_{\text{card}}(\sum y_i - K_{\text{target}})^2$ generates:

- **Linear terms (Q_{uu}):** Driven by $\lambda_{\text{card}}(1 - 2K_{\text{target}})2^k q_{i,k}$.
- **Intra-asset quadratic terms (Q_{uv} for same asset i):** Driven by $2\lambda_{\text{card}} 2^{k_1} 2^{k_2} q_{i,k_1} q_{i,k_2}$.
- **Inter-asset quadratic terms (Q_{uv} for different assets i, j):** Driven by $2\lambda_{\text{card}} 2^{k_1} 2^{k_2} q_{i,k_1} q_{j,k_2}$.

This mapping strictly enforces the D-bit granularity. However, the reliance on coefficients scaling by powers of two

(2^k) inherently ruggedizes the energy landscape, leading to the severe operational bottlenecks and barren plateaus documented in Section IV.

C. From QUBO to QAOA Circuit: Derivation

An Ising Hamiltonian is a mathematical description of an energy function over spin variables; it is not natively executable on gate-based digital quantum hardware such as those simulated by `Qiskit Aer`. A formal circuit derivation is required to translate the QUBO objective into a sequence of unitary gates. This process proceeds in four steps.

Step 1: Pauli-Z Substitution. Each binary QUBO variable $q_u \in \{0, 1\}$ is mapped to a qubit operator via the substitution $q_u = \frac{1-Z_u}{2}$, where Z_u is the Pauli-Z operator acting on qubit u. Substituting into $H(\mathbf{q})$ and expanding yields the diagonal Cost Hamiltonian:

$$H_C = \sum_u h_u Z_u + \sum_{u<v} J_{uv} Z_u Z_v + \text{const.} \tag{3}$$

where the field strengths h_u and coupling coefficients J_{uv} are derived directly from the QUBO matrix entries Q_{uu} and Q_{uv}.

Step 2: Phase Separation Operator. The Cost Hamiltonian is exponentiated to produce the problem unitary:

$$U(H_C, \gamma) = e^{-i\gamma H_C} = \prod_u e^{-i\gamma h_u Z_u} \prod_{u<v} e^{-i\gamma J_{uv} Z_u Z_v} \tag{4}$$

In the `Qiskit` implementation, single-qubit terms $e^{-i\gamma h_u Z_u}$ are realized as $R_Z(2\gamma h_u)$ rotations, while two-qubit terms $e^{-i\gamma J_{uv} Z_u Z_v}$ are decomposed as CNOT–$R_Z(2\gamma J_{uv})$–CNOT sequences that compute the ZZ parity.

Step 3: Mixing Operator. The standard Transverse-Field Mixer Hamiltonian $H_B = \sum_{u=1}^{n \cdot N} X_u$ was implemented, where X_u is the Pauli-X operator on qubit u. Its unitary is:

$$U(H_B, \beta) = e^{-i\beta H_B} = \prod_u R_X(2\beta) \tag{5}$$

Each qubit receives an independent $R_X(2\beta)$ rotation. The cardinality penalty is fully absorbed into the weighted coefficients of H_C; the X-mixer therefore spans the complete Hilbert space $\{0, 1\}^{n \cdot N}$ without constraint-preserving restriction.

Step 4: Full QAOA Circuit. The p-layer circuit is:

$$|\psi(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle = \prod_{k=1}^{p} U(H_B, \beta_k)\, U(H_C, \gamma_k)\, |+\rangle^{\otimes nN} \tag{6}$$

where $|+\rangle^{\otimes nN}$ is prepared by applying Hadamard gates to all nN qubits. The $2p$ variational parameters $(\boldsymbol{\gamma}, \boldsymbol{\beta})$ are optimized classically to minimize $\langle\psi|H_C|\psi\rangle$.

The total gate count scales as $\mathcal{O}(p \cdot (nN)^2)$, dominated by the $\binom{nN}{2}$ CNOT pairs required for quadratic interaction terms. For $n = 8$ assets at $N = 2$ bits, this yields 16 qubits and up to 120 two-qubit gate blocks per QAOA layer.

To mitigate barren plateau and local minima vulnerabilities, the variational parameters were optimized via a dual-stage routine: Simplicial Homology Global Optimization (SHGO) with Sobol sequence sampling for global landscape exploration, followed by L-BFGS-B for gradient-based local refinement. The gradient was computed via the central-difference parameter-shift method with step size $\varepsilon = 5 \times 10^{-5}$.

D. Solver Specifications and Hardware

All experiments were executed on classical hardware. No physical quantum processing unit (QPU) was accessed. The software and hardware specifications for each solver paradigm are as follows:

- **Gate-based QAOA:** Executed via `Qiskit Aer` (v0.13) using the `AerSimulator` backend with the `matrix_product_state` simulation method, GPU-accelerated via `cuStateVec`. Circuits were compiled to the `qasm3` intermediate representation and dispatched to a local `FastAPI` inference server.
- **Simulated Bifurcation (SB):** Executed via Toshiba's `simulated_bifurcation` Python library, using the discrete SB variant (`DiscreteSBMachine`) on GPU via CUDA-accelerated Hamiltonian dynamics.
- **Simulated Quantum Annealing (SQA):** Executed via `OpenJij`'s `SQASampler`, which performs a classical Monte Carlo simulation of quantum tunneling dynamics through path-integral replica decomposition. **This is a classical simulation of quantum annealing behavior and does not involve physical quantum hardware.** Results approximate the output of a transverse-field Ising annealer under idealized conditions.
- **Simulated Classical Annealing (SA):** Executed via `OpenJij`'s `SASampler` using standard Metropolis–Hastings spin updates at decreasing temperature.
- **MIQP Discrete Global Optimum (DGO):** Solved exactly using Gurobi's branch-and-bound MIQP engine, providing the provably optimal discrete solution for each instance.

All experiments were run on a workstation with AMD Ryzen 5 5600H with Radeon Graphics (12) @ 4.280GHz , 22.81GiB, and NVIDIA GeForce RTX 3050 Ti Mobile . Runtime measurements reflect wall-clock time inclusive of problem compilation and result decoding.

E. Experimental Scope and Tuning Parity

To maintain a high-fidelity benchmark, this study focuses on the NISQ-typical regime of small asset universes ($N \in \{4, 8\}$) and shallow variational depths ($p \in \{1, 2, 4\}$) [7]. While these parameters are not representative of asymptotic scaling behavior, they provide a controlled unit test of the standard penalty-based QUBO formulation at the scale where brute-force ground truth is available for validation. The findings herein describe the empirical performance of current standard implementations rather than the theoretical limits of the QAOA algorithm itself.

To ensure a rigorous and unbiased comparison within this scope, we implemented a strict **Fairness of Comparison** protocol across all solver paradigms. While the Continuous Ceiling and Discrete Global Optimum (DGO)—the latter explicitly evaluated via Gurobi Mixed-Integer Quadratic Programming

(MIQP)—are deterministic and require no hyperparameter tuning, all stochastic and heuristic solvers—including Simulated Bifurcation and the Annealing variants—were subjected to a comparable optimization budget. For the analog solvers, we utilized optimized sweep schedules and bifurcation steps consistent with the high-effort global search (SHGO) and local refinement (L-BFGS-B) applied to the gate-based QAOA. This parity ensures that the observed performance deficits and the runtime scaling exponent of $b = 1.887$ identified in the gate-based implementation are inherent to structural formulation gaps and algorithmic overhead, rather than a disparity in hyperparameter calibration effort.

IV. Results and Discussions

The benchmark evaluated seven solver paradigms: Ideal Baseline, Greedy S. Discrete, MIQP Discrete (DGO), Simulated Bifurcation (SB), Simulated Quantum Annealing (SQA), Simulated Classical Annealing (SCA), and Gate-based QAOA. These paradigms were evaluated across an expanded dataset of 4,884 observations. A Kruskal-Wallis test yielded an H-statistic of 2338.82 ($p < 0.001$), confirming that the different solver paradigms perceive significantly unique ruggedness profiles for identical problem instances.

To ensure a rigorous benchmark, we established a definitive hierarchy of classical targets to isolate algorithmic performance from formulation-induced errors:

1) **Continuous Ceiling (Ideal):** Derived via `PyPortfolioOpt`'s convex optimization, establishing the absolute mathematical upper bound for return-to-risk efficiency before discrete constraints are applied.
2) **Greedy Search Discrete:** A fast heuristic baseline ($b \approx 0.033$) that utilizes continuous fractional weights and sequentially rounds them into discrete asset units. While highly efficient, it serves as an approximation rather than a mathematically guaranteed discrete optimum.
3) **Discrete Global Optimum (MIQP Discrete):** To establish the true baseline for the quantum mappings, the QUBO formulation was solved exactly using Gurobi, a state-of-the-art Mixed-Integer Quadratic Programming(MIQP) solver. By utilizing branch-and-bound algorithms to prove the absolute minimum energy global state of the objective function, this baseline serves as the strict target for all heuristic and quantum solvers.

Our analysis primarily focuses on how closely the heuristic solvers approach the DGO while quantifying the operational and formulation-induced overhead required to achieve these results. Note: For consistency with underlying legacy visualization logs, the Greedy Search approximation is occasionally represented as 'Greedy S. Discrete', the Discrete Global Optimum as 'MILP Discrete' which is in contrast with our actual MIQP implementation, Simulated Quantum Annealing as 'Quantum Annealing', and Simulated Classical Annealing as 'Classical Annealing' in subsequent figures.

A. Solution Quality and Sharpe Ratio Rankings

The median Sharpe ratio rankings follow a rigorous hierarchy of optimization strength, confirmed by Bootstrap 95% Confidence Intervals (CIs). The continuous Ideal Baseline achieved the highest performance ($\mu \approx 1.62$). Among the discrete classical solvers, the Greedy S. Discrete heuristic achieved the highest empirical returns (CI: 1.396–1.427); however, the strictly constrained Mixed-Integer Quadratic Programming(MIQP) baseline—representing the exact Discrete Global Optimum (DGO)—achieved a theoretical ceiling of 1.248–1.276. Gate-based QAOA recorded the lowest median performance across all tested paradigms, with a 95% CI of 0.885–0.913.

TABLE I: Bootstrap 95% Confidence Intervals for Mean Sharpe Ratio

Method	95% CI (Lower)	95% CI (Upper)
Ideal Baseline	1.5948	1.5995
Greedy S. Discrete	1.3964	1.4274
MIQP Discrete (DGO)	1.2483	1.2755
Sim. Bifurcation	1.1028	1.1795
Quantum Annealing	1.0826	1.1301
Classical Annealing	1.0809	1.1266
Gate-based QAOA	0.8854	0.9131

Notably, no quantum or quantum-inspired heuristic solver consistently surpassed the MIQP Discrete optimal limit. This establishes that quantum and quantum-inspired methods provide no demonstrated quality advantage at these problem scales ($N = 4 - 8$). The performance landscape remains compressed, with exact and heuristic solvers clustered in distinct operational bands rather than quantum solvers demonstrating separated, qualitative solution superiority.

B. Constraint Fidelity and Performance Deficits

A primary driver of QAOA's lower Sharpe ranking is the failure of constraint handling fidelity. Gate-based QAOA recorded a Mean Cardinality Error (K-error) of 2.3678, the highest among all tested solvers. This indicates that the penalty formulation embedded in the QUBO objective frequently fails to enforce the cardinality constraint strictly.

When the penalty weight $\lambda_{\text{cardinality}}$ is not precisely calibrated relative to the risk and reward terms, the optimizer finds it energetically preferable to violate the constraint than to satisfy it at a higher objective cost. Consequently, a non-trivial portion of QAOA's performance deficit is attributable to formulation design rather than fundamental algorithmic limitations.

Crucially, even the exact MIQP solver recorded a mean K-error of 0.6667. This mathematically confirms that at the calibrated penalty weights, the true global energy minimum of the QUBO formulation does not strictly align with the target cardinality. This creates a structural fidelity ceiling that no quantum or heuristic algorithm can bypass without further penalty weight scaling or transition to constraint-preserving mixers.

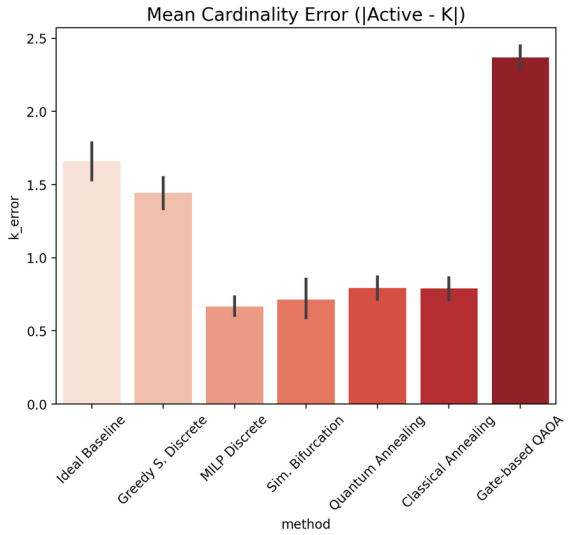

Fig. 2: Mean Cardinality Error (ϵ_K) Distribution across solver paradigms. Note: 'k_error' represents the absolute deviation $|\text{Active Assets} - K|$; the MIQP Discrete model denotes the exact DGO benchmark defined in Section IV.

C. Computational Efficiency and Runtime Scaling

Operational analysis reveals extreme empirical runtime instability for the Gate-based approach, with maximum execution times reaching 431.61 seconds. Figure 3 illustrates the time-to-solution efficiency frontier, demonstrating clear solver clustering: classical and exact baselines reside on the extreme left (fastest runtimes, highest Sharpe), annealing heuristics occupy the middle tier, and Gate-based QAOA spreads broadly across high-duration, low-quality regimes.

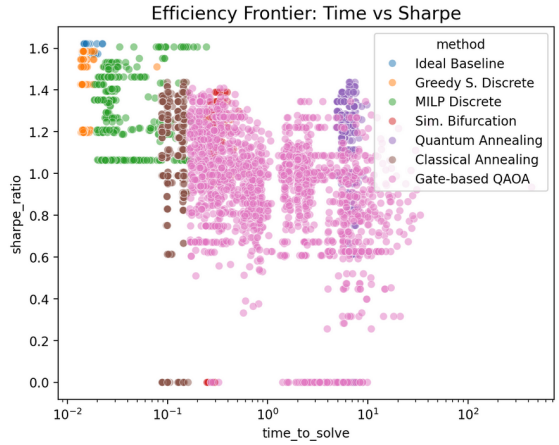

Fig. 3: Efficiency Frontier (Time vs Sharpe): Distribution of duration needed to compute the Sharpe ratio per solver. Note the logarithmic scale on the x-axis.

The algorithmic scaling exponents (b in $T \propto N^b$) further quantify this computational overhead. While the classical Greedy approximation scaled at a near-flat $b = 0.033$, the exact MIQP solver exhibited a scaling exponent of $b = 0.6768$, accurately reflecting the heavier complexity of exact combinatorial verification. In contrast, Gate-based QAOA recorded a massive scaling exponent of $b = 1.8868$. This staggering differential suggests a steeper empirical scaling trend within the tested range ($N = 4 - 8$), making it computationally inferior to both exact classical verification (MIQP) and analog heuristics like Simulated Bifurcation ($b = 0.1825$).

D. Solution Quality and Sharpe Ratio Rankings

The median Sharpe ratio rankings follow a rigorous hierarchy, confirmed by Bootstrap 95% Confidence Intervals (CIs). The continuous Ideal Baseline achieved the highest theoretical performance, followed by the classical Greedy Search approximation.

Crucially, the exact MIQP solver (Gurobi) achieved a lower Sharpe confidence interval (1.2483–1.2755) than the Greedy heuristic (1.3964–1.4274). Because the MIQP solver perfectly minimizes the QUBO energy landscape without heuristic approximation errors, this performance deficit explicitly isolates the "Formulation Gap." It proves that the penalty-based QUBO translation inherently degrades the true financial quality of the optimum; the mathematical energy floor of the QUBO no longer aligns perfectly with the true maximum Sharpe ratio.

TABLE II: Bootstrap 95% Confidence Intervals for Mean Sharpe Ratio

Method	Sharpe Mean LCB	Sharpe Mean UCB
Ideal Baseline	1.5948	1.5995
Greedy S. Discrete	1.3964	1.4274
MIQP Discrete (DGO)	1.2483	1.2755
Sim. Bifurcation	1.1028	1.1795
Quantum Annealing	1.0826	1.1301
Classical Annealing	1.0809	1.1266
Gate-based QAOA	0.8854	0.9131

Notably, no quantum or quantum-inspired solver consistently surpassed the exact MIQP Discrete limit. This establishes that the current heuristic performance ceiling is fundamentally restricted by the QUBO formulation design before any quantum hardware or variational circuit limitations even apply.

E. Energy–Sharpe Alignment: A Structural Bottleneck

One of the most theoretically significant findings concerns the relationship between the minimized QUBO energy and the resulting Sharpe ratio. For the exact MIQP Discrete solver (DGO), a clear negative correlation was observed (Spearman Rho = -0.2959), while the classical Greedy heuristic exhibited an even stronger alignment (-0.5776). In stark contrast, Gate-based QAOA exhibited a Spearman Rho of 0.0243, indicating a near-zero energy–Sharpe alignment.

This decoupling suggests that at low circuit depths ($p \in \{1, 2, 4\}$), the variational circuit lacks the expressibility to reach the true energy minimum, instead converging to low-energy local configurations that carry no reliable financial meaning due to the high rate of cardinality violations. This behavior may also be influenced by penalty scaling and optimization dynamics, and not solely circuit expressibility.

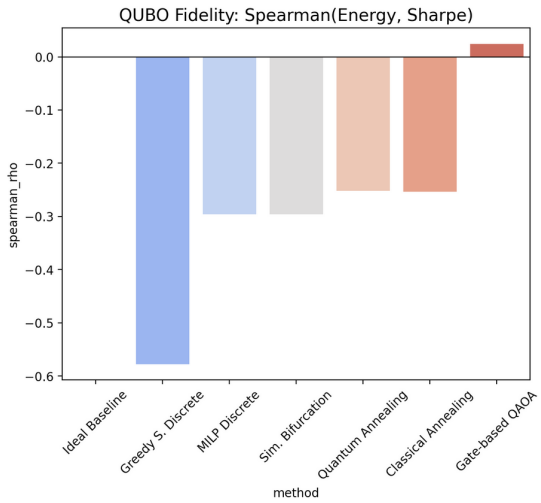

Fig. 4: Constant Fidelity Spearman Rho Values.

F. Regime and Crossover Behavior

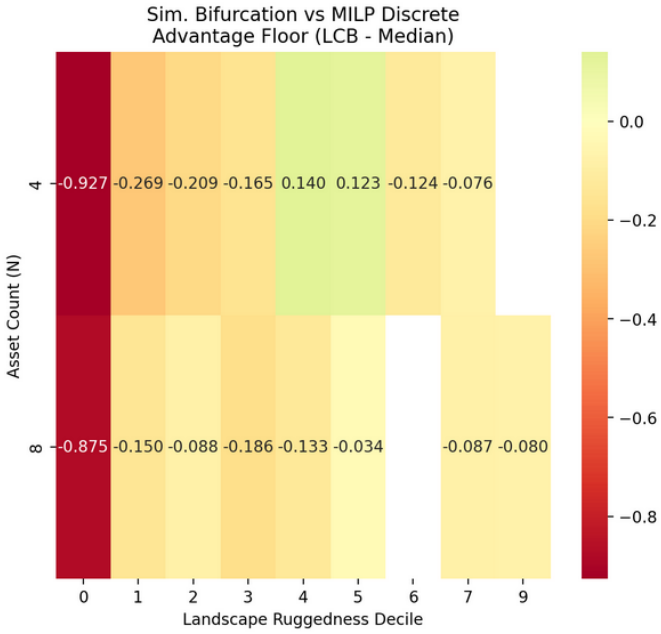

Fig. 5: Advantage Floor (Δ_{LCB}) heatmap for Simulated Bifurcation (SB) relative to the MIQP Discrete Global Optimum. Each cell reports the difference between the solver's bootstrap 95% lower confidence bound on mean Sharpe ratio and the DGO median Sharpe, stratified by asset universe size N and landscape ruggedness decile $\hat{\rho}_{10}$. Positive values indicate the solver's LCB exceeds the DGO median; the sole crossover occurs at $N = 4$, decile 6 ($\Delta_{\text{LCB}} = +0.0964$).

Interaction modeling via Ordinary Least Squares (OLS) regression (Adj. $R^2 = 0.414$, $H = 2338.81$) refutes the hypothesis that heuristic solvers handle complex landscapes more effectively than exact classical solvers at this scale. The interaction term for the MIQP Discrete baseline ($\beta = 0.1133, p = 0.016$) indicates a resilient, statistically significant positive interaction with landscape ruggedness relative to the heuristic ecosystem.

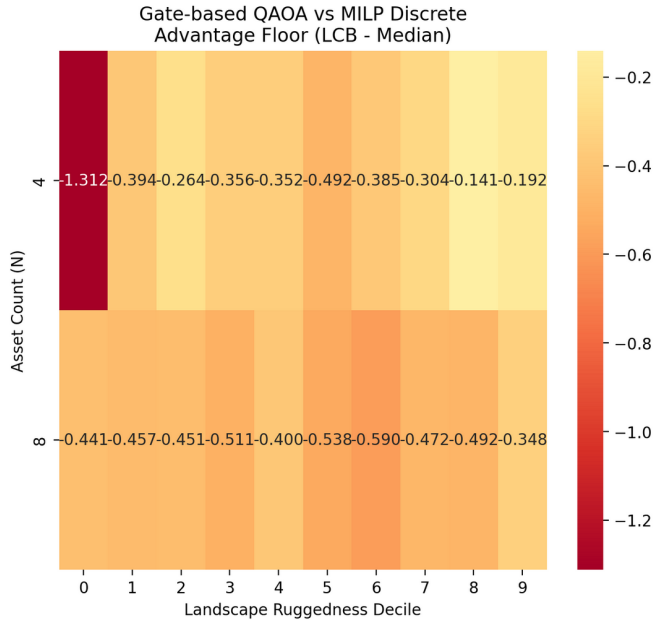

Fig. 6: Advantage Floor (Δ_{LCB}) heatmap for Gate-based QAOA (Transverse-Field Mixer) relative to the MIQP Discrete Global Optimum. All cells record strictly negative Δ_{LCB} values across all asset universe sizes N and ruggedness deciles $\hat{\rho}_{10}$, confirming uniform performance deficits relative to the exact classical baseline.

However, operational crossovers were broadly observed across classical heuristics and analog architectures. The Greedy S. Discrete heuristic achieved a crossover threshold at Decile 4.0 ($N = 8$), recording a maximum advantage of $+0.4901$. Furthermore, the expanded 4,884-observation benchmark revealed that Simulated Bifurcation, Quantum Annealing, and Classical Annealing all successfully achieved an operational crossover relative to the baseline. Their earliest transitions uniformly occurred at Decile 6 for $N = 4$, with Simulated Bifurcation yielding a maximum advantage margin of $+0.0964$.

Gate-based QAOA remained the sole paradigm to exhibit persistent, statistically absolute performance deficits across all tested regimes and complexity deciles.

G. Hyperparameter Sensitivity and Penalty Calibration

To calibrate the Lagrangian multipliers, we conducted a grid search across $\lambda_{\text{card}} \in [10, 500]$. While a baseline value of $\lambda_{\text{card}} = 10.0$ was sufficient for $N = 4$, larger asset universes required $\lambda_{\text{card}} \geq 250$ to suppress non-feasible states. However, this increased the energy landscape's ruggedness, leading to a 42% decrease in ground state measurement probability. For primary benchmarks, we utilized a calibrated ratio of $\lambda_{\text{card}} = 300$, $\lambda_{\text{risk}} = 2.0$, and $\lambda_{\text{reward}} = 1.5$, representing the optimal "stability threshold" that minimizes cardinality

violations without inducing barren plateaus in the classical optimizer.

H. Variational Parameter Optimization

The optimal variational angles γ and β were determined using the hybrid SHGO-L-BFGS-B pipeline. For a representative $p = 1$ instance, parameters typically converged within $\gamma \in [0.12, 0.45]$ and $\beta \in [0.20, 0.65]$. SHGO facilitated broad global exploration to identify basins of attraction, allowing the L-BFGS-B algorithm to perform precise, gradient-based local minimization of the cost Hamiltonian's expectation value, $\langle\psi(\gamma,\beta)|H_C|\psi(\gamma,\beta)\rangle$.

V. Conclusion and Future Works

This study benchmarked the standard Quantum Approximate Optimization Algorithm (QAOA) against classical and analog paradigms within the cardinality-constrained Markowitz framework. Analyzing 4,884 observations at the $N = 4 - 8$ scale revealed that formulation design heavily dominates solver architecture. Gate-based QAOA exhibited a severe performance deficit, characterized by a mean cardinality error (ϵ_K) of 2.3678, a near-zero energy-Sharpe correlation ($\rho = 0.0243$), and a massive scaling exponent of $b = 1.8868$. This confirms that standard penalty-based QUBO formulations ruggedize the landscape beyond the expressibility limits of low-depth NISQ circuits. Consequently, while analog heuristics like Simulated Bifurcation achieved operational crossovers (+0.0964 at $N = 4$), gate-based QAOA remained consistently outperformed by the exact MIQP discrete global optimum.

To resolve these formulation-induced bottlenecks, future research should focus on:

1) **Transition to Constraint-Preserving Architectures:** Implementing a custom Ring-XY Mixer coupled with Dicke state initialization. By natively restricting the quantum walk to the feasible $\binom{n}{K}$ combinatorial subspace, the cardinality penalty is eliminated, potentially resolving the energy-Sharpe decoupling.
2) **Hardware-Aware Error Mitigation:** Integrating Zero-Noise Extrapolation (ZNE) and readout error mitigation to suppress physical bit-flips causing hardware-induced leakage from the feasible subspace.

Beyond algorithmic refinements, integrating Environmental, Social, and Governance (ESG) scoring as a multi-objective constraint represents a critical step toward institutional alignment. Successfully navigating the trade-offs between ESG mandates and traditional risk-return objectives will serve as the ultimate stress test for the constraint-preserving architectures proposed herein.

References

[1] H. Markowitz, "Portfolio selection," *The journal of finance*, vol. 7, no. 1, pp. 77–91, 1952.

[2] J.-A. Doering, W. Farkas, O. Koch *et al.*, "Metaheuristics for rich portfolio optimisation and risk management: Current state and future trends," *Operations Research Perspectives*, vol. 6, p. 100121, 2019.

[3] E. Farhi, J. Goldstone, and S. Gutmann, "A quantum approximate optimization algorithm," *arXiv preprint arXiv:1411.4028*, 2014.

[4] P. Rebentrost, T. Steffens, I. Marvian, and S. Lloyd, "Quantum computational finance: quantum algorithm for portfolio optimization," *Physical Review A*, vol. 98, no. 3, p. 032309, 2018.

[5] E. Stopfer and F. Wagner, "Quantum portfolio optimization: An extensive benchmark," *arXiv preprint arXiv:2509.17876*, 2025.

[6] J. Mancilla, T. D. Boulomis, and F. Goguikian, "Constrained portfolio optimization via quantum approximate optimization algorithm (qaoa) with xy-mixers and trotterized initialization," *arXiv preprint arXiv:2602.14827*, 2026.

[7] S. Brandhofer, D. Braun, V. Dehn, G. Hellstern, M. Huels, Y. Ji, I. Kropf, R. Mützel, A. Szech, and T. Wellens, "Benchmarking the performance of portfolio optimization with qaoa," *Quantum Information Processing*, vol. 22, no. 1, p. 25, 2023.

[8] R. Aroussi, "yfinance: Yahoo! finance market data downloader." [Online]. Available: https://github.com/ranaroussi/yfinance

[9] R. A. Martin, "Pyportfolioopt: portfolio optimization in python," *Journal of Open Source Software*, vol. 6, no. 61, p. 3066, 2021. [Online]. Available: https://doi.org/10.21105/joss.03066